# ECONOMIC IMPACT REPORT

**Provenance-Backed Securities and the Washington State Economy**

*Market Projections, Job Creation, GDP Contribution, and Sector-by-Sector Analysis*

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## Executive Summary

This Economic Impact Report presents a comprehensive analysis of the market opportunity, job creation potential, GDP contribution, and sector-by-sector implications of the Washington Provenance-Backed Securities Act. The findings demonstrate that Washington State stands at a unique intersection of technological infrastructure, agricultural heritage, research excellence, and regulatory opportunity that positions it to capture first-mover advantage in defining the ownership infrastructure for intangible assets—the dominant form of value in the 21st-century economy.

### ★ Headline Economic Findings

$97.6T

Global Corporate Intangible Asset Value (2025)

92%

S&P 500 Market Cap in Intangible Assets

$18.9T

Projected Tokenized RWA Market by 2033

8,500–34,000

Projected Jobs Created in WA by Year 10

$3.2B–$14.8B

Projected 10-Year Cumulative PBS Transaction Volume

Intangible assets now constitute approximately 92% of S&P 500 market capitalization, yet no standardized, liquid marketplace exists for the vast majority of these assets—particularly genetic intellectual property, accumulated human expertise, AI training data provenance, and performance-verified creative works. The Provenance-Backed Securities framework addresses this structural gap. Existing real-world asset tokenization projections (BCG/Ripple: $18.9 trillion by 2033) largely exclude intangible IP, meaning PBS targets an additional, untapped layer of global value.

Under moderate assumptions, PBS operations could contribute $580 million to $1.4 billion annually to Washington's GDP by Year 10, create 14,000–22,000 direct and indirect jobs, and generate $120–$280 million in annual state revenue through B&O taxes, transaction fees, and licensing. Washington's no-income-tax structure, world-class technology and agricultural sectors, and premier university research infrastructure make it the ideal jurisdiction to establish this framework—provided it acts before competing states or nations claim the first-mover position.

# Section 1: The Global Intangible Asset Market Opportunity

## 1.1 The Intangible Economy Inversion

The global economy has undergone what Ocean Tomo defines as an "economic inversion"—a wholesale transformation in the nature of value creation whereby economic worth has migrated from what can be "touched" to what can be

“thought.” The Ocean Tomo 2025 Intangible Asset Market Value Study, released in February 2026, documents this transformation across 50 years of U.S. market data:

- **1975:** Tangible assets represented 83% of S&P 500 market value; intangible assets accounted for only 17%.
- **2025:** Intangible assets now constitute approximately **92%** of S&P 500 market capitalization, while tangible assets have been reduced to a mere 8%.
- This 75-percentage-point shift occurred within a single human lifespan, with particularly rapid acceleration between 1985 and 2005 when intangible asset market value surged from 32% to 79%.
- Notably, S&P 500 intangible asset values remained stable at approximately 90% through 2020–2025 despite the most aggressive monetary tightening cycle in four decades—demonstrating structural resilience.

## 1.2 Global Intangible Asset Valuation

According to the WIPO Global Innovation Index and the Brand Finance Global Intangible Finance Tracker (GIFT) 2025, the value of corporate intangible assets worldwide approaches **$97.6 trillion** as of 2025—close to $100 trillion. Key findings include:

- Global corporate intangible value grew 23% from 2024 to 2025, following a 28% surge in 2024.
- This represents a 16-fold increase since 1996.
- Over the past decade, global intangible asset value has averaged approximately 67% of global GDP.
- The United States leads as the most intangible-asset-intensive economy, with its top 15 firms’ intangible assets comprising 78% of total enterprise value.
- Eight of the world’s top 10 companies by intangible value are U.S.-based, including NVIDIA ($4.3 trillion), Microsoft, Apple, Amazon, and Alphabet.

## 1.3 The Tokenization Wave—and Its Critical Gap

A landmark April 2025 report by Boston Consulting Group and Ripple projects the market for tokenized real-world assets to grow from approximately $0.6 trillion today to **$18.9 trillion by 2033**, representing a compound annual growth rate of 53%. The projection identifies a three-phase evolution:

1. **Phase 1 – Low-Risk Adoption:** Institutions tokenize familiar instruments like money market funds and bonds.
2. **Phase 2 – Institutional Expansion:** Scaling into complex assets such as private credit and real estate ($6.8 trillion projected).
3. **Phase 3 – Market Transformation:** Tokenization becomes embedded in financial and non-financial products.

> ⚠ **The Critical Gap PBS Addresses**
>
> The BCG/Ripple $18.9 trillion projection covers mutual funds, bonds, real estate, alternative investments, and commodities—but it **largely excludes** intangible IP assets such as genetic intellectual property, accumulated human expertise, AI training data provenance, scientific research outputs, and performance-verified creative works. PBS targets this additional, untapped layer of global value that sits *on top of* the existing tokenization opportunity.

## 1.4 The “Dead Capital” Analogy

Peruvian economist Hernando de Soto estimated that 5.3 billion people globally hold approximately **$9.3 trillion in “dead capital”**—real estate and physical assets that are informally held, lack legal recognition, and cannot be converted into financial instruments. De Soto argued that this dead capital exists because the formal property systems exclude most of the world’s population from converting physical assets into fungible capital.

The intangible economy faces an analogous—and far larger—dead capital problem. While global corporate intangible assets approach $100 trillion, the vast majority of intangible value created by *individuals*—seed breeders, independent inventors, creative workers, data generators, manufacturing process experts, researchers—remains "dead." This individual intangible capital lacks:

- Standardized documentation and provenance verification
- Liquid marketplaces for exchange
- Performance-tracking mechanisms (oracles)
- Legal frameworks recognizing individual ownership
- Tradeable securities instruments

If de Soto's $9.3 trillion in dead real estate capital justifies systemic property reform, then the far larger pool of dead intangible capital—potentially an order of magnitude greater—justifies the creation of Provenance-Backed Securities infrastructure.

# Section 2: Washington State Economic Baseline

*Washington State possesses a uniquely favorable combination of economic characteristics for PBS implementation: a technology-driven economy, world-class agricultural production, premier research universities, and a tax structure that makes PBS transaction fee revenue especially attractive.*

## 2.1 Key Economic Indicators

| Indicator | Value (2025) |
|---|---|
| Real GDP | $717.5 billion (2025); 2.2% growth from 2024 |
| GDP Per Capita | $89,700 (2025) |
| Civilian Labor Force | ~4.07 million (March 2026) |
| Nonfarm Employment | ~3.64 million (seasonally adjusted) |

| Indicator | Value (2025) |
|---|---|
| Unemployment Rate | 5.1% (March 2026) |
| Agricultural Production Value | ~$14.0 billion (2023 record; 300+ crops grown) |
| Top GDP Sector: Information | $159.7 billion (2025) |
| Professional & Business Services GDP | $98.9 billion (2025) |
| UW Total R&D Expenditures (2024) | $1.69 billion (Ranked #8 nationally) |
| State Income Tax | None (revenue from B&O tax and sales tax) |

## 2.2 Key Sector Strengths

**Technology:** Washington is home to Microsoft (Redmond), Amazon (Seattle), and major offices for Google, Meta, and numerous cloud and AI companies. The Information sector alone contributed $159.7 billion to state GDP in 2025, making it the single largest industry contributor. This technology concentration provides the workforce, infrastructure, and corporate demand environment essential for PBS implementation.

**Agriculture:** Washington ranks among the top agricultural states nationally, with total production value reaching a record $14.0 billion in 2023. The state produces over 300 different crops, leads the nation in apple production (67% of U.S. supply), and is a major producer of hops, cherries, potatoes, wheat, and wine grapes. This agricultural diversity makes Washington the ideal pilot jurisdiction for SeedBid, the first planned PBS deployment.

**Aerospace:** Boeing’s extensive manufacturing presence and the broader aerospace supply chain represent significant concentrations of manufacturing process IP and engineering expertise—key PBS domains.

**Clean Energy and Maritime:** Washington’s hydropower resources, clean energy sector, and maritime industry provide additional sectors where process IP and technical innovation create PBS-addressable intangible value.

## 2.3 Revenue Structure Advantage

Washington’s lack of a personal or corporate income tax creates a revenue structure that relies primarily on Business and Occupation (B&O) tax and sales tax.

This structure makes PBS-generated revenue streams—transaction fees, registration fees, examination fees, and B&O tax on exchange operations—an especially attractive mechanism for revenue diversification without requiring new tax legislation.

## 2.4 University Research Infrastructure

The University of Washington (UW) ranks **8th nationally** in total R&D expenditures at $1.69 billion (2024), positioning it among the most research-intensive universities in the country. Washington State University (WSU) adds significant research capacity, particularly in agricultural science, veterinary medicine, and the IMPACT Center for agricultural competitiveness. This research infrastructure generates substantial intangible IP that currently lacks liquid market access—a gap PBS directly addresses.

# Section 3: PBS Market Size Projections—Three Scenarios

*The following projections model PBS transaction volume, job creation, and revenue generation across conservative, moderate, and aggressive adoption scenarios over a 10-year horizon. All figures are denominated in 2026 dollars.*

## 3.1 Conservative Scenario

**Assumptions:** SeedBid agricultural pilot only in Years 1–2, targeting 3–5% of Washington's agricultural IP value. Gradual expansion to 3 domains by Year 5 (adding creative works and tech patents). By Year 10, 5 domains active with modest national participation through a federal framework. Limited interstate adoption.

| Year | Active Domains | Annual PBS Transaction Volume | Cumulative Volume | Direct Jobs | Total Jobs (Direct + Indirect) |
|---|---|---|---|---|---|
| 1 | 1 (Agriculture) | $45 million | $45 million | 65 | 120 |
| 2 | 1 | $120 million | $165 million | 110 | 220 |

| Year | Active Domains | Annual PBS Transaction Volume | Cumulative Volume | Direct Jobs | Total Jobs (Direct + Indirect) |
|---|---|---|---|---|---|
| 3 | 2 | $240 million | $405 million | 200 | 450 |
| 4 | 3 | $380 million | $785 million | 320 | 780 |
| 5 | 3 | $480 million | $1.27 billion | 450 | 1,200 |
| 6 | 4 | $580 million | $1.85 billion | 580 | 1,800 |
| 7 | 4 | $650 million | $2.50 billion | 680 | 2,400 |
| 8 | 5 | $720 million | $3.22 billion | 780 | 3,200 |
| 9 | 5 | $790 million | $4.01 billion | 880 | 4,400 |
| 10 | 5 | $860 million | $4.87 billion | 1,000 | 5,800 |

**Conservative 10-Year Totals:** $4.87 billion cumulative transaction volume; 1,000 direct jobs; 5,800 total jobs (including indirect and induced); ~$48 million annual state revenue by Year 10.

## 3.2 Moderate Scenario

**Assumptions:** SeedBid launch plus early creative works pilot in Years 1–2. Five domains active by Year 3–5 with interstate compact involving 3–5 states and federal bill enacted by Year 4. Full 8-domain operation by Year 8 with Pacific Rim international exchange initiated by Year 9.

| Year | Active Domains | Annual PBS Transaction Volume | Cumulative Volume | Direct Jobs | Total Jobs (Direct + Indirect) |
|---|---|---|---|---|---|
| 1 | 2 | $95 million | $95 million | 120 | 250 |
| 2 | 3 | $280 million | $375 million | 250 | 580 |
| 3 | 5 | $620 million | $995 million | 520 | 1,400 |
| 4 | 5 | $1.1 billion | $2.10 billion | 850 | 2,600 |
| 5 | 6 | $1.6 billion | $3.70 billion | 1,200 | 4,200 |
| 6 | 7 | $2.1 billion | $5.80 billion | 1,650 | 6,400 |
| 7 | 8 | $2.7 billion | $8.50 billion | 2,100 | 9,200 |
| 8 | 8 | $3.2 billion | $11.70 billion | 2,600 | 12,400 |
| 9 | 8 + International | $3.9 billion | $15.60 billion | 3,200 | 16,800 |

| Year | Active Domains | Annual PBS Transaction Volume | Cumulative Volume | Direct Jobs | Total Jobs (Direct + Indirect) |
|---|---|---|---|---|---|
| 10 | 8 + International | $4.6 billion | $20.20 billion | 3,800 | 22,000 |

**Moderate 10-Year Totals:** $20.2 billion cumulative transaction volume; 3,800 direct jobs; 22,000 total jobs; ~$185 million annual state revenue by Year 10.

## 3.3 Aggressive Scenario

**Assumptions:** Multi-domain launch from Year 1 (agriculture, creative works, patents). Federal framework enacted by Year 3 with 6+ domains active. Washington established as national PBS hub by Year 5. International PBS exchange headquarters established in Washington by Year 6, with full global integration by Year 8–10.

| Year | Active Domains | Annual PBS Transaction Volume | Cumulative Volume | Direct Jobs | Total Jobs (Direct + Indirect) |
|---|---|---|---|---|---|
| 1 | 3 | $180 million | $180 million | 220 | 480 |
| 2 | 5 | $580 million | $760 million | 520 | 1,300 |
| 3 | 6 | $1.4 billion | $2.16 billion | 1,100 | 3,200 |
| 4 | 7 | $2.8 billion | $4.96 billion | 1,800 | 5,800 |
| 5 | 8 (National Hub) | $4.5 billion | $9.46 billion | 2,800 | 10,200 |
| 6 | 8 + Intl. HQ | $6.2 billion | $15.66 billion | 3,600 | 14,800 |
| 7 | 8 + Global | $8.1 billion | $23.76 billion | 4,500 | 19,600 |
| 8 | 8 + Global | $10.4 billion | $34.16 billion | 5,600 | 24,400 |
| 9 | 8 + Global | $12.8 billion | $46.96 billion | 6,800 | 29,800 |
| 10 | 8 + Full Global | $15.2 billion | $62.16 billion | 8,200 | 34,000 |

**Aggressive 10-Year Totals:** $62.16 billion cumulative transaction volume; 8,200 direct jobs; 34,000 total jobs; ~$420 million annual state revenue by Year 10.

# Section 4: Job Creation Analysis

## 4.1 Direct Job Categories

| Job Category | Roles | Year 5 Range | Year 10 Range | Avg. Salary Estimate |
|---|---|---|---|---|
| WPBSC Regulatory Staff | Commissioners, examiners, compliance officers, legal counsel, administrative staff | 15–25 | 25–40 | $95,000–$165,000 |
| PBS Exchange Operators | Platform engineers, operations managers, customer support, market makers | 150–600 | 400–2,400 | $110,000–$195,000 |
| Oracle Developers & Data Scientists | Performance oracle design, data integration, ML/AI engineers, domain specialists | 80–350 | 250–1,500 | $135,000–$220,000 |
| Blockchain Engineers | Smart contract developers, security auditors, protocol engineers | 60–250 | 180–1,200 | $140,000–$230,000 |
| Compliance Professionals | KYC/AML analysts, regulatory reporting, compliance officers | 40–180 | 120–800 | $90,000–$160,000 |
| Domain-Specific Specialists | Agricultural IP evaluators, patent analysts, creative works appraisers, data provenance auditors | 100–400 | 300–2,000 | $85,000–$175,000 |

## 4.2 Indirect and Induced Jobs

**Indirect jobs** include positions created in supporting industries:

- Legal services specializing in PBS law, IP valuation, and securities compliance
- Accounting and audit firms conducting PBS-specific financial reviews
- Consulting firms advising businesses on PBS integration and strategy
- Technology infrastructure providers (cloud services, cybersecurity, data centers)
- Agricultural technology integration firms bridging SeedBid to existing farm operations
- Academic programs and training providers developing PBS workforce pipelines

**Induced jobs** result from the spending multiplier effects of new high-wage employment. PBS positions average well above the Washington median household income of approximately $90,000, with many technical roles exceeding $150,000. The Bureau of Economic Analysis financial services multiplier of approximately 2.5–3.2x suggests that each direct PBS job supports 1.5–2.2 additional indirect and induced positions.

## 4.3 Comparative Benchmark: Wyoming Blockchain Initiative

Wyoming provides a useful—if limited—comparison case. Since 2018, Wyoming has enacted 30+ blockchain-related laws, including the world’s first token taxonomy, Special Purpose Depository Institution (SPDI) charters, DAO LLC recognition, and a FinTech Sandbox. Wyoming’s blockchain-friendly regulatory environment has attracted company formations and relocations, including Kraken Financial (first SPDI charter) and numerous DAO entities. The University of Wyoming now offers a blockchain minor. Wyoming continues to pursue asset-backed digital token initiatives through its Select Committee on Blockchain, Financial Technology and Digital Innovation Technology.

PBS represents a more comprehensive and higher-value proposition than Wyoming's blockchain framework, as it creates entirely new asset classes rather than simply providing regulatory clarity for existing digital assets. Washington's far larger economy ($717.5 billion GDP vs. Wyoming's ~$47 billion) and existing technology workforce provide a dramatically larger base for job creation.

## 4.4 Total Job Creation Summary

| Metric | Conservative | Moderate | Aggressive |
|---|---|---|---|
| Direct Jobs by Year 5 | 450 | 1,200 | 2,800 |
| Total Jobs by Year 5 | 1,200 | 4,200 | 10,200 |
| Direct Jobs by Year 10 | 1,000 | 3,800 | 8,200 |
| Total Jobs by Year 10 | 5,800 | 22,000 | 34,000 |
| Average Salary (Direct PBS Roles) | $115,000–$185,000 (benchmarked against WA tech sector median of ~$145,000) | | |

# Section 5: GDP and Revenue Contribution

## 5.1 Direct GDP Contribution

PBS exchange operations, WPBSC administrative functions, and the supporting professional services ecosystem contribute directly to Washington's GDP through employee compensation, business revenue, and capital investment. Using Bureau of Economic Analysis methodology and standard financial services sector multipliers (2.5–3.2x), the following GDP contributions are projected:

| Metric | Conservative | Moderate | Aggressive |
|---|---|---|---|
| Annual GDP Contribution, Year 5 | $180 million | $580 million | $1.4 billion |
| Annual GDP | $420 million | $1.4 billion | $4.2 billion |

| Metric | Conservative | Moderate | Aggressive |
|---|---|---|---|
| Contribution, Year 10 | | | |
| GDP Contribution as % of WA GDP (Year 10) | 0.06% | 0.19% | 0.59% |

## 5.2 State Revenue Projections

Washington State would derive revenue from PBS operations through four primary mechanisms:

| Revenue Source | Mechanism | Year 5 (Moderate) | Year 10 (Moderate) |
|---|---|---|---|
| B&O Tax on PBS Exchange Revenue | Standard B&O rate applied to exchange gross receipts | $18 million | $52 million |
| PBS Transaction Fees (0.1–0.5%) | Per-transaction fee on PBS trades, collected by WPBSC | $48 million | $115 million |
| Registration & Licensing Fees | Exchange operator licensing, domain registration, professional certifications | $4 million | $8 million |
| Examination & Filing Fees | PBS issuance examination fees, annual compliance filings | $6 million | $12 million |
| **Total Annual State Revenue** | | **$76 million** | **$187 million** |

## 5.3 Revenue Comparison Across Scenarios

| Timeframe | Conservative | Moderate | Aggressive |
|---|---|---|---|
| Year 5 Annual Revenue | $28 million | $76 million | $165 million |
| Year 10 Annual Revenue | $48 million | $187 million | $420 million |

| Timeframe | Conservative | Moderate | Aggressive |
|---|---|---|---|
| 5-Year Cumulative Revenue | $72 million | $195 million | $440 million |
| 10-Year Cumulative Revenue | $265 million | $910 million | $2.28 billion |

For context, Washington's total state revenue exceeds $60 billion annually. While PBS revenue represents a small percentage of the total, it constitutes a *new* revenue stream derived from an entirely new economic sector, requires no new taxes or tax rate increases, and grows with market adoption rather than population or consumption.

# Section 6: Sector-by-Sector Impact Analysis

*The following analysis examines each of the eight PBS domains through the lens of Washington State's specific economic strengths, identifying the addressable market opportunity, key stakeholders, and projected impact for each sector.*

## 6.1 Seed Genetics and Agriculture

| Parameter | Washington Context |
|---|---|
| Current WA Agricultural Value | ~$14.0 billion total production (2023 record); 300+ crops; 31,800 farm operations |
| PBS Addressable Opportunity | Seed genetics IP, variety development rights, breeding program outputs; estimated $350M–$700M addressable within WA |
| Key Stakeholders | Independent plant breeders, WSU agricultural research, organic seed producers, hop and apple variety developers, tribal agricultural programs |
| Market Concentration Problem | Four firms (Bayer, Corteva, Syngenta, BASF) control 56% of the global commercial seed market and 75% of GE seed sales. U.S. corn seed prices have risen 344% since 2000 ($66/acre to $293/acre in 2024). The |

| Parameter | Washington Context |
|---|---|
| | Herfindahl-Hirschman Index (HHI) for U.S. corn seed exceeds 3,000—well above the DOJ/FTC threshold of 2,500 indicating severe concentration. |

**Impact:** SeedBid, the first PBS deployment, creates a transparent, performance-verified marketplace for seed genetics. Independent breeders gain liquid market access for varieties currently locked behind proprietary licensing walls. Performance oracles track yield data, disease resistance, and climate adaptability, enabling evidence-based price discovery that breaks the information asymmetry favoring large agrochemical corporations. Washington's diverse agricultural sector—spanning apples, hops, wine grapes, wheat, potatoes, and specialty crops—provides an ideal testing ground across multiple crop categories.

## 6.2 Creative Works

| Parameter | Washington Context |
|---|---|
| WA Creative Economy | Major gaming hub: Nintendo of America (Redmond), Valve (Bellevue), Bungie (Bellevue). Music heritage: Sub Pop Records, vibrant independent music scene. Growing film production sector. |
| PBS Addressable Opportunity | Fractional ownership of creative IP with performance-oracle-tracked royalties, streaming revenue, and audience engagement metrics; estimated $200M–$500M addressable within WA |
| Key Stakeholders | Independent game developers, musicians, visual artists, writers, podcasters, content creators, small studios |

**Impact:** PBS enables fractional ownership of creative works with transparent, oracle-verified performance tracking. An independent musician can issue PBS instruments backed by verified streaming revenue. A small game studio can raise capital against documented engagement metrics. This disrupts the current model where creative value is captured almost entirely by platform intermediaries and major labels/publishers.

## 6.3 Patents and Technical IP

| Parameter | Washington Context |
|---|---|
| WA Patent Activity | Washington generates thousands of patents annually. Microsoft, Amazon, and Boeing are among the world's most prolific patent holders. Dense tech corridor from Seattle to Redmond to Bellevue. |
| PBS Addressable Opportunity | Secondary market for patent licensing rights with performance data; estimated $500M–$1.5B addressable within WA |
| Key Stakeholders | Independent inventors, small tech companies, university tech transfer offices, Boeing supply chain innovators |

**Impact:** PBS creates a liquid secondary market for patent licensing rights. Independent inventors gain access to capital markets previously available only to large corporations with dedicated licensing departments. Performance oracles track licensing revenue, citation impact, and commercial deployment, enabling transparent price discovery for technical IP.

## 6.4 Personal Data

| Parameter | Washington Context |
|---|---|
| Global Data Broker Market | ~$278 billion (2024), projected to reach $512–$525 billion by 2033 at ~7.3% CAGR. North America holds 41.2% market share. |
| WA Legislative Precedent | Washington's My Health My Data Act (2023) established state-level precedent for individual data ownership and consent requirements—one of the strongest data privacy laws in the nation. |
| PBS Addressable Opportunity | Individual data ownership with provenance verification; enables consumers to participate in data economy value currently captured by brokers; estimated $150M–$400M addressable within WA |

**Impact:** PBS enables individuals to own, provenance-document, and trade their personal data through verified securities instruments. Instead of data brokers

extracting and reselling consumer data without compensation, PBS creates a framework where individuals participate as asset owners. Washington's My Health My Data Act demonstrates the state's willingness to lead on data sovereignty—PBS extends this principle from passive protection to active economic participation.

## 6.5 Scientific Research

| Parameter | Washington Context |
|---|---|
| UW Research Expenditures | $1.69 billion in total R&D (2024), ranked 8th nationally among all U.S. universities |
| Key Research Strengths | UW: biomedical sciences, computer science, oceanography, clean energy. WSU: agricultural science, veterinary medicine, food science. Fred Hutchinson Cancer Center. |
| PBS Addressable Opportunity | Researcher ownership stakes in publicly funded discoveries; commercialization pathway for university IP; estimated $200M–$600M addressable within WA |

**Impact:** PBS enables researchers to retain ownership stakes in discoveries funded by public grants, creating a direct commercialization pathway for university IP. Performance oracles track citation impact, licensing activity, and downstream commercial applications. This addresses the longstanding tension between publicly funded research and private commercialization by ensuring researchers participate in the value their work generates.

## 6.6 Manufacturing Processes

| Parameter | Washington Context |
|---|---|
| WA Manufacturing Sector | ~269,000 manufacturing jobs (2025). Boeing's aerospace manufacturing. Advanced materials, clean tech manufacturing, food processing. |
| PBS Addressable Opportunity | Documentation and valuation of accumulated manufacturing process knowledge; estimated $100M–$300M addressable within WA |

| Parameter | Washington Context |
|---|---|
| Key Stakeholders | Experienced manufacturing workers, process engineers, aerospace supply chain companies, food processing specialists |

**Impact:** PBS documents and values accumulated manufacturing knowledge—the "tribal knowledge" that exists in experienced workers' expertise but is rarely formally recognized or compensated. When automation systems or AI are trained on process expertise developed by workers over decades, PBS ensures those workers retain ownership of and receive compensation for their contributed knowledge. This is particularly significant in Boeing's aerospace supply chain, where process precision expertise represents enormous intangible value.

## 6.7 Human Capital

| Parameter | Washington Context |
|---|---|
| WA Labor Force | ~4.07 million workers across all sectors |
| AI Displacement Context | Washington's tech concentration means both high AI development activity and significant worker exposure to AI displacement. The AI training data market is growing rapidly, with companies using worker-generated expertise to train systems that may ultimately replace those workers. |
| PBS Addressable Opportunity | Every worker's accumulated expertise as provenance-documented, performance-verified, tradeable assets; AI training royalties; largest long-term PBS domain |

### ★ The Core Meritocratic Proposition

Human Capital PBS is the most transformative domain. Every Washington worker's accumulated expertise becomes a provenance-documented, performance-verified, tradeable asset. When a worker's expertise is used to train an AI system, PBS ensures that worker receives ongoing royalties from the value their knowledge generates. This converts the existential threat of AI

displacement into an economic opportunity: workers become *asset owners* rather than *displaced labor*. The meritocratic principle that value should flow to those who create it is codified in securities law.

## 6.8 Biopharmaceutical

| Parameter | Washington Context |
|---|---|
| WA Biotech Sector | Seattle biotech cluster includes major research institutions, clinical trial networks, and pharmaceutical companies. Fred Hutchinson Cancer Center is a world leader in immunotherapy research. |
| PBS Addressable Opportunity | Fractional ownership of drug development pipelines, clinical trial data, genetic research outputs; performance oracles tracking clinical outcomes; estimated $300M–$800M addressable within WA |
| Key Stakeholders | Biotech startups, university researchers, clinical research organizations, patient advocacy groups, independent research scientists |

**Impact:** PBS enables fractional ownership of drug development pipelines with performance oracles tracking clinical trial outcomes and regulatory milestones. Small biotech firms and individual researchers gain access to capital markets for their IP without surrendering full ownership to pharmaceutical conglomerates. Performance oracles provide transparent tracking of clinical phase progression, efficacy data, and commercialization revenue.

# Section 7: Competitive Positioning—First-Mover Analysis

## 7.1 State-Level Innovation Legislation Comparison

| State | Legislative Action | Scope | Economic Outcome |
|---|---|---|---|
| Wyoming | 30+ blockchain laws since 2018; SPDI charter; DAO LLC recognition; Stable Token Act; FinTech Sandbox; Chancery Court | Regulatory clarity for existing digital assets; new financial institution types; business formation | Attracted company formations and relocations; University of Wyoming blockchain minor; positioned as regulatory leader for digital assets despite small economy (~$47B GDP) |
| Colorado | Digital Token Act (2019) | Exemptions for utility tokens from state securities laws under certain conditions | Provided regulatory clarity for token issuers; positioned Colorado as crypto-friendly; limited broader economic restructuring |
| Arizona | FinTech Sandbox (2018) | Regulatory sandbox allowing fintech companies to test products with reduced regulatory requirements | Accepted dozens of participants; provided testing environment; limited to sandbox scope |
| Utah | Regulatory Sandbox (2019) | Broader regulatory sandbox extending beyond fintech to multiple industries | Enabled testing of innovative products; modest economic footprint relative to state economy |
| **Washington (Proposed)** | **Provenance-Backed Securities Act** | **Creation of entirely new asset classes; comprehensive ownership infrastructure for intangible assets; 8-domain** | **Projected: 5,800–34,000 jobs; $265M–$2.28B cumulative state revenue; potential "Delaware effect" for intangible asset law** |

| State | Legislative Action | Scope | Economic Outcome |
|---|---|---|---|
| | | **framework; dedicated regulatory commission** | |

## 7.2 The “Delaware Effect”

Delaware dominates American corporate law not because of any inherent geographic advantage, but because it **moved first** to create specialized legal infrastructure—the Court of Chancery (1792), the Delaware General Corporation Law, and a body of case law that provides predictability and expertise. Today, 67.8% of Fortune 500 companies are incorporated in Delaware, generating over $2 billion annually in franchise tax revenue—more than a quarter of the state’s total revenue.

Washington has the opportunity to establish a comparable “Delaware effect” for intangible asset ownership law. By creating the WPBSC, establishing the PBS regulatory framework, and building the body of administrative law and expertise around intangible asset securities, Washington can become the default jurisdiction for PBS issuance, exchange, and dispute resolution—attracting businesses, legal talent, and regulatory expertise from across the nation and eventually the world.

## 7.3 International Competition

Washington’s window is not unlimited. Several international jurisdictions are aggressively pursuing digital asset and tokenization frameworks:

- **Singapore:** The Monetary Authority of Singapore (MAS) has established a comprehensive regulatory framework for digital payment tokens and is actively piloting tokenized asset infrastructure.
- **Switzerland:** The Swiss DLT Act provides legal recognition for tokenized securities, and the Swiss Financial Market Supervisory Authority (FINMA) has established clear regulatory guidelines.

- **United Kingdom:** The UK is developing its digital securities sandbox and has signaled intent to become a global crypto-asset hub.
- **European Union:** The Markets in Crypto-Assets (MiCA) regulation provides EU-wide framework for digital assets.
- **United Arab Emirates:** Dubai has established active tokenized real estate markets and regulatory infrastructure.

However, none of these jurisdictions have addressed provenance-backed securities for intangible IP assets specifically. The opportunity to define this category remains open—but it will not remain open indefinitely.

## 7.4 Risk of Inaction

If Washington does not act, the consequences are concrete and measurable:

- Another state establishes the PBS framework first, capturing the headquarters, jobs, and regulatory authority
- International jurisdictions attract PBS exchange operations, routing transaction fee revenue offshore
- Washington’s technology workforce—ideally positioned for PBS roles—migrates to jurisdictions with PBS infrastructure
- The “Delaware effect” compounds: once a jurisdiction establishes legal infrastructure and case law, the switching costs become prohibitive
- Washington’s agricultural IP community (independent breeders, WSU researchers) continues to lack liquid market access while other states enable it

# Section 8: Risk Factors and Sensitivity Analysis

## 8.1 Risk Matrix

| Risk Factor | Probability | Impact | Overall Rating | Mitigation Strategy |
|---|---|---|---|---|
| Market Adoption Risk | Medium | High | Medium | Phased domain rollout beginning with SeedBid agricultural pilot where demand is demonstrated; education and outreach programs; regulatory sandbox for early adopters |
| Regulatory Uncertainty (Federal) | Medium-High | High | Medium-High | Proactive federal engagement strategy (Document 4); design PBS framework for compatibility with existing SEC rules; interstate compact to build coalition for federal bill |
| Technology Risk | Low-Medium | Medium | Low-Medium | Blockchain and oracle technology is maturing rapidly; leverage existing infrastructure (Ethereum, Chainlink); technology-agnostic statutory |

| Risk Factor | Probability | Impact | Overall Rating | Mitigation Strategy |
|---|---|---|---|---|
| | | | | language |
| Competition Risk | Medium | High | Medium | First-mover advantage through rapid legislative action; establish WPBSC as center of expertise; build network effects before competitors enter |
| Legal Challenge Risk | Medium | Medium | Medium | Industry incumbents (Big Ag, Big Tech) may challenge through litigation or lobbying; robust constitutional analysis (Document 3); preemption-resistant design |
| Valuation Methodology Risk | Medium | Medium | Medium | Performance oracles provide objective, verifiable valuation data; WPBSC establishes standardized valuation frameworks; domain-specific advisory committees |

## 8.2 Sensitivity Analysis

The following table shows how changes in key assumptions affect Moderate Scenario Year-10 projections:

| Variable Changed | -50% | -25% | Base Case | +25% | +50% |
|---|---|---|---|---|---|
| Adoption Rate | $2.3B vol. / 11,000 jobs | $3.5B vol. / 16,500 jobs | $4.6B vol. / 22,000 jobs | $5.8B vol. / 27,500 jobs | $6.9B vol. / 33,000 jobs |
| Transaction Volume per Domain | $2.3B vol. / $94M rev. | $3.5B vol. / $140M rev. | $4.6B vol. / $187M rev. | $5.8B vol. / $234M rev. | $6.9B vol. / $280M rev. |
| Fee Rate (0.1–0.5% range) | $57M rev. | $94M rev. | $187M rev. | $234M rev. | $280M rev. |
| Domain Expansion Timeline | 6 domains by Yr 10 | 7 domains by Yr 10 | 8 domains + Intl by Yr 10 | 8 + 2 Intl by Yr 10 | 8 + 3 Intl by Yr 10 |

**Key finding:** Even under the most pessimistic sensitivity assumptions (all variables at -50%), PBS operations still generate approximately $57 million in annual state revenue and support approximately 5,500 total jobs by Year 10—well exceeding the break-even threshold for the state's initial investment in WPBSC establishment.

# Section 9: Strategic Recommendations

*The following phased implementation roadmap is designed to maximize economic impact while managing risk through controlled expansion.*

## Phase 1: Foundation (Years 1–2)

- **SeedBid Agricultural Pilot Launch:** Deploy the first PBS exchange targeting Washington seed genetics, demonstrating the concept with a sector where need and state expertise are well-established.
- **WPBSC Establishment:** Stand up the Washington Provenance-Backed Securities Commission with initial regulatory staff (15–25 positions), rulemaking authority, and examination capabilities.

- **Regulatory Sandbox:** Launch a PBS regulatory sandbox to enable controlled testing of exchange platforms, oracle systems, and compliance frameworks.
- **University Partnerships:** Engage UW and WSU to develop PBS-related academic programs, research initiatives, and workforce pipeline development.
- **Estimated Investment:** $8–$15 million in state establishment costs, offset by initial registration and licensing fees.

## Phase 2: Expansion (Years 2–4)

- **Creative Works Domain:** Launch PBS exchange for creative works, leveraging Washington's gaming, music, and content creation sectors.
- **Patent/Technical IP Domain:** Activate patent licensing PBS, engaging the tech corridor's patent portfolios and independent inventors.
- **Interstate Compact Negotiations:** Initiate negotiations with 3–5 states for mutual recognition of PBS instruments and exchange licensure.
- **Federal Bill Advocacy:** Coordinate with congressional delegation to introduce PBS federal framework legislation (Document 4).

## Phase 3: National Integration (Years 4–6)

- **Human Capital Domain:** Launch the transformative Human Capital PBS domain, enabling AI training royalties and expertise securitization.
- **Personal Data Domain:** Activate personal data PBS, building on My Health My Data Act precedent.
- **Federal Framework Enactment:** Support passage of federal PBS legislation positioning Washington as the administrative center.
- **National PBS Exchange Hub:** Establish Washington as the primary PBS exchange operations center.

## Phase 4: Global Leadership (Years 6–10)

- **Full 8-Domain Operation:** All PBS domains active and mature.
- **International PBS Exchange:** Establish international PBS exchange headquarters in Washington, beginning with Pacific Rim market integration (Japan, South Korea, Singapore, Australia).
- **Global Standard-Setting:** Position WPBSC as the global standard-setter for intangible asset securities regulation.
- **Continuous Innovation:** Support emergence of new PBS domains as the intangible economy evolves.

### ★ Washington's Window of Opportunity

First-mover advantage in defining intangible asset ownership infrastructure is a **once-in-a-generation economic opportunity** comparable to Delaware's establishment of corporate law dominance. The global intangible economy approaches $100 trillion. The tokenization wave is accelerating. AI is rapidly transforming the relationship between human expertise and economic value. The jurisdiction that establishes the legal, regulatory, and market infrastructure for Provenance-Backed Securities will capture the headquarters, the jobs, the revenue, the expertise, and the compounding network effects for decades to come.

Washington possesses every prerequisite: the technology workforce, the agricultural heritage, the research universities, the legislative capacity, and the economic scale. What it requires now is the **political will to act before another jurisdiction claims this position**.

*The window is open. It will not stay open forever.*

*Data sources include: Ocean Tomo/J.S. Held IAMV Study (Feb. 2026); WIPO/Brand Finance GIFT 2025; BCG/Ripple RWA Tokenization Report (Apr. 2025); USDA/NASS; Bureau of Economic Analysis; Bureau of Labor Statistics; U.S. NSF/NCSES; Washington State Employment Security Department; Grand View Research; ETC Group/GRAIN (2025). All projections are modeled estimates and subject to the assumptions stated herein.*